

Data Pipeline Architecture and Functionality

Introduction

A data pipeline is a process that automates the flow of data from a source to a destination through multiple stages such as extraction, transformation, and loading. This project implements a simple ETL (Extract, Transform, Load) pipeline using Python.

Pipeline Architecture

The pipeline consists of three main stages. The first stage is data extraction, where raw data is loaded from a CSV file. The second stage is data transformation, where missing values are handled and basic cleaning operations are performed. The final stage is data loading, where the processed data is saved into a new file.

Functionality

The extract function reads raw data from the source file. The transform function cleans the data by removing null values and standardizing column names. The load function saves the cleaned dataset into a destination file. The entire process is executed in sequence using the `run_pipeline` function.

Conclusion

This pipeline demonstrates how raw data can be automatically processed into a clean format. It helps in understanding the basic concept of data pipelines and how ETL processes work in real-world applications.